PERSONAL PORTFOLIO / VERSION 1.0

Airbnb Market Lab

Albany, New York | Snapshot 2026-09-14

A personal data engineering and market intelligence project

LISTINGS	HOSTS	REVIEW RECORDS
500	200	30,251

From source files to defensible decisions. This project ingests a real public snapshot, checks its quality, builds a dimensional warehouse, explores market structure and tests how well listing attributes predict published prices.

The purpose is practical learning and a reviewable portfolio: reproducible code, explicit assumptions, visible failures and evidence-backed communication. No hiring or assessment outcome is claimed.

Headline: Median positive listed price is $144.00; 33 listings have no usable positive price. Calendar prices are absent in the source.

Source: Inside Airbnb, CC BY 4.0. Independent project; not affiliated with Airbnb. Prices are in the configured local currency (USD for Albany).

HOW TO REVIEW THIS PROJECT

Reading guide

Start with the executive brief for the business story, then read the quality and modeling sections to understand what supports it. The repository contains executable evidence behind every result.

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Companion artifacts: explorer.html for filters; analysis.json for statistical detail; schema_profile.json for source fields; quality_report.json for validation counts; sql/analytics.sql for warehouse queries.

RESEARCH & ENGINEERING

Executive brief

The snapshot contains 500 listings operated by 200 observed hosts. Of those listings, 467 have a positive published price. The median is $144.00, while the mean is $292.59; the difference makes segment-specific comparisons preferable to a single city average.

The largest 10% of observed hosts control 44.2% of listings. This is a useful concentration signal for supply monitoring. It does not identify business ownership: one account may represent multiple owners, and one owner may use several accounts.

The mean listing-level unavailable share is 30.9%. Calendar state does not reveal the reason for unavailability. No realised occupancy, booking growth or income figure is inferred from it.

Decision	Evidence	Practical next step
Build comparable groups	Prices vary with property mix	Filter by room type, capacity and ward before comparing prices.
Track missing prices	Some listings lack a usable price	Keep missing-price supply visible and avoid median imputation in descriptive reports.
Use model errors as a guardrail	Evaluation separates hosts	Treat predictions as experiments until repeat snapshots validate stability.
Delay income claims	Calendar pricing is unavailable here	Obtain verified booking and transaction data before estimating realised returns.

The most valuable result is a repeatable evidence trail. Every headline can be traced to source hashes, cleaning rules, analytical tables and generated artifacts.

RESEARCH & ENGINEERING

Scope and success criteria

The project focuses on one market so that ingestion, validation, modeling and communication can be tested together. Albany was already selected in the starter repository. Its manageable size supports a complete local run without paid services.

Implemented workstream	What completion means
Source understanding	All seven configured files downloaded; tabular schemas profiled; source dictionary and caveats reviewed.
Data engineering	Verified raw cache, deterministic cleaning, referential checks, DuckDB star schema and run metadata.
Analysis	Price/supply/review/availability summaries, statistical tests, confidence intervals and regression diagnostics.
Prediction	Three model families plus a median baseline; host-separated holdout and group cross-validation.
Communication	Offline explorer, generated report, notebook, SQL examples and a learning guide.

Success means a reviewer can reproduce the results and identify what the data cannot answer. It does not mean a production pricing service, a complete census of all rentals or a causal evaluation of host behavior.

Cross-city work, demand forecasting, NLP, LLM integration and cloud deployment are deliberately deferred. Those extensions require additional validation, source history or evaluation labels. Adding them without those prerequisites would dilute the reliability of the core project.

The supplied brief informed the topic selection. The original confidential document is not copied into project artifacts or source control.

RESEARCH & ENGINEERING

Source inventory and semantics

Input	Rows / features	Grain and purpose
listings	500	One source listing; detailed attributes and price
calendar	182,500	Listing and calendar date; future availability
reviews	30,251	Review ID; listing reference and publication date
listings_summary	500	Summary listing attributes; reconciliation input
reviews_summary	30,251	Listing and date; repeated same-day reviews are valid
neighbourhoods	16	Ward names and optional group
boundaries	17	Ward polygons for geographic context

The archive label is 2026-09-14, while the observed listing scrape date is 2026-09-16. Tenure and recent-review windows use the observed scrape date; the archive label remains the partition key.

Identifiers are strings throughout ingestion and cleaning. Some listing IDs exceed the exact integer range of common floating-point formats. Converting them through floats would silently break joins.

A host is an account, a listing is an advertised accommodation unit, and a review is a published guest review. Summary review rows have no unique review ID; repeated listing/date pairs are not automatically duplicates.

Source dictionary notes: price is in local currency; host registration date may predate hosting; geographic coordinates are approximate. Newer account-duration fields are interpreted as years plus months/12 when host_since is absent; this is an explicit schema assumption, not an exact reconstructed date.

RESEARCH & ENGINEERING

Data quality: evidence first

Check	Observed result	Treatment
Missing listing price	33	Keep listing; exclude from price statistics/models.
Negative listing price	0	Replace invalid price with null; report the count.
Duplicate listing keys	0	Deterministically retain last input record.
Calendar orphan/invalid rows	0	Remove before warehouse publication.
Review orphan/invalid/future rows	0	Remove and count; retained source files permit diagnosis.
Calendar coverage	365-365 days/listing	Expose coverage; denominators use known states.
Review-count reconciliation	0	Compare detailed review counts to listing totals.
IQR price outliers	20	Flag and retain; unusual is not necessarily wrong.

schema_profile.json covers row counts, null fractions, cardinality, inferred types and safe example values for every CSV column. Source files are read as strings first, so inference is descriptive rather than an implicit conversion policy.

Required-column failures stop the pipeline. Optional-field gaps produce nulls or an explicit unavailable analysis. A successful run means the configured checks passed, not that source values are objectively true.

Quality metrics are not hidden cleanup details: they determine which business questions remain answerable.

RESEARCH & ENGINEERING

Cleaning decisions and duplicates

Currency strings are parsed with an explicit export format: a leading currency symbol, optional comma thousands separators and a decimal point. Malformed strings are not guessed. Negative prices become null, while zero prices remain visible but are excluded from positive-price comparisons.

Dates use a fixed ISO format. Geographic/rating/availability domains are checked. 500 bathroom counts are recovered from explicit text such as '1.5 shared baths'; ambiguous text remains null. 500 account tenures use source years/months because registration dates are absent. Zero/missing bedrooms do not produce price-per-bedroom values.

Exact duplicate keys use a stable keep-last policy in source order. This is operationally reproducible, but it is not a claim that the final duplicate is the most accurate. A source revision can be compared using the archived raw hashes.

The title-similarity heuristic flagged 43 candidate pairs within host and room/capacity blocks. A similarity score of at least 0.92 identifies records for inspection. Separate units often have nearly identical names, so no fuzzy candidate is automatically deleted.

Field family	Policy	Reason
Categorical labels	Trim, normalize whitespace/case	Avoid accidental groups caused by formatting.
Missing numeric values	Retain null in analytical tables	Preserve uncertainty and unbiased denominators.
ML predictors	Training-fold median / mode imputation	Prevent leakage from held-out records.
Public outputs	Exclude identity/text/coordinates	Share aggregate evidence without republishing personal profiles.

Fuzzy comparisons are capped at 200 records per block and can miss renamed units or cross-account duplicates. This heuristic is a review aid, not an entity-resolution system.

RESEARCH & ENGINEERING

Architecture and lineage

Layer	Artifact	Transformation
Source	Inside Airbnb dated URLs	Pinned city/snapshot config; seven files.
Raw	data/raw/city/date	Retry downloads; validate gzip; hash each file.
Extracted	data/extracted/city/date	Recreate CSV/GeoJSON cache from verified raw bytes.
Validated	In-memory typed frames	Key checks, domain checks, deduplication, quality audit.
Warehouse	data/warehouse/city/date.duckdb	Atomic database replacement after assertions.
Analysis	outputs/city/date/analysis.json	Summaries, tests, regression and model evaluation.
Presentation	Figures, explorer and this report	Generated from analysis results; no manual headline edits.

A run fingerprint combines source checksums, city configuration, package code and dependency versions. The pipeline skips unchanged work only when every recorded output still matches its hash. A missing or damaged artifact forces regeneration.

The database is built in a temporary sibling file and moved into place after primary-key constraints and orphan assertions pass. Source downloads are promoted from partial files only after length and gzip integrity checks.

Report files are written during the run, and the success manifest is published last. Consumers should use only runs marked successful. Full cross-file transactional publication and concurrent-writer locking are future hardening work.

> Lineage rule: no analysis reads directly from an unverified network response. Raw bytes remain available locally for every result.

RESEARCH & ENGINEERING

Dimensional model and SQL

Table	Primary key	Meaning
dim_listing	id	Listing snapshot plus review/calendar features.
dim_host	host_id	Observed account portfolio in this city.
dim_neighbourhood	neighbourhood	Observed ward label.
dim_date	date	Dates occurring in calendar/reviews.
fact_calendar	listing_id + date	Availability, optional price and stay rules.
fact_review	id	Listing reference and review date only.
snapshot_metadata	single row	City, archive date and currency.

fact_calendar.listing_id and fact_review.listing_id join to dim_listing.id. Listing rows refer to host and neighbourhood dimensions; both facts refer to dim_date. Primary keys are enforced in DuckDB; referential integrity is asserted during publication.

Each database contains one city snapshot. Its effective grain includes the city and archive date encoded in the database path. This prevents overwriting history and avoids treating a review repeated in later snapshots as a newly written review.

The enriched listing dimension intentionally contains derived measures for convenient exploration. A larger warehouse would split slowly changing listing attributes from snapshot facts and use explicit compound or surrogate keys.

Two views expose ward metrics and monthly calendar summaries. sql/analytics.sql demonstrates price comparison, supply concentration, availability denominators and review activity. A date spine would be preferable for zero-filled time series in a production BI layer.

DuckDB was chosen for local analytical SQL with no server setup. SQLite was viable for storage; PostgreSQL adds operational overhead at this scale.

RESEARCH & ENGINEERING

Repeatability and operation

Start with Python 3.11 or later, create a virtual environment, install the pinned dependencies, and run the command-line pipeline. The default config pins Albany rather than silently switching to a newer release.

Command	Purpose
python -m airbnb_lab.cli ingest	Download or validate source cache.
python -m airbnb_lab.cli run	Build all analytical outputs and report.
python -m airbnb_lab.cli run --offline	Rebuild or skip using only verified local inputs.
python -m airbnb_lab.cli run --force --offline	Regenerate even when inputs and code match.
python -m pytest -q	Run synthetic-data correctness tests.

Incremental behavior is snapshot-level: a new date creates a separate raw partition, warehouse and output folder. Unchanged snapshots skip work. Row-level CDC, deletion tracking and Type 2 dimension intervals are not implemented.

Download retries use bounded exponential delay. Failures leave an error/status record and exit nonzero. The CI workflow runs unit and integration tests against synthetic data without depending on public downloads.

Docker Compose supplies a reproducible local execution path and persistent data/output mounts. It is a delivery configuration, not evidence of a cloud deployment. The environment verification notes identify whether Docker was executable on the authoring machine.

For a second city, supply its official path, date, currency and source file map. The code accepts configurations, but this release has been validated on Albany only. Numeric comparisons across currencies must not be pooled without a justified conversion design.

RESEARCH & ENGINEERING

Price structure and comparisons

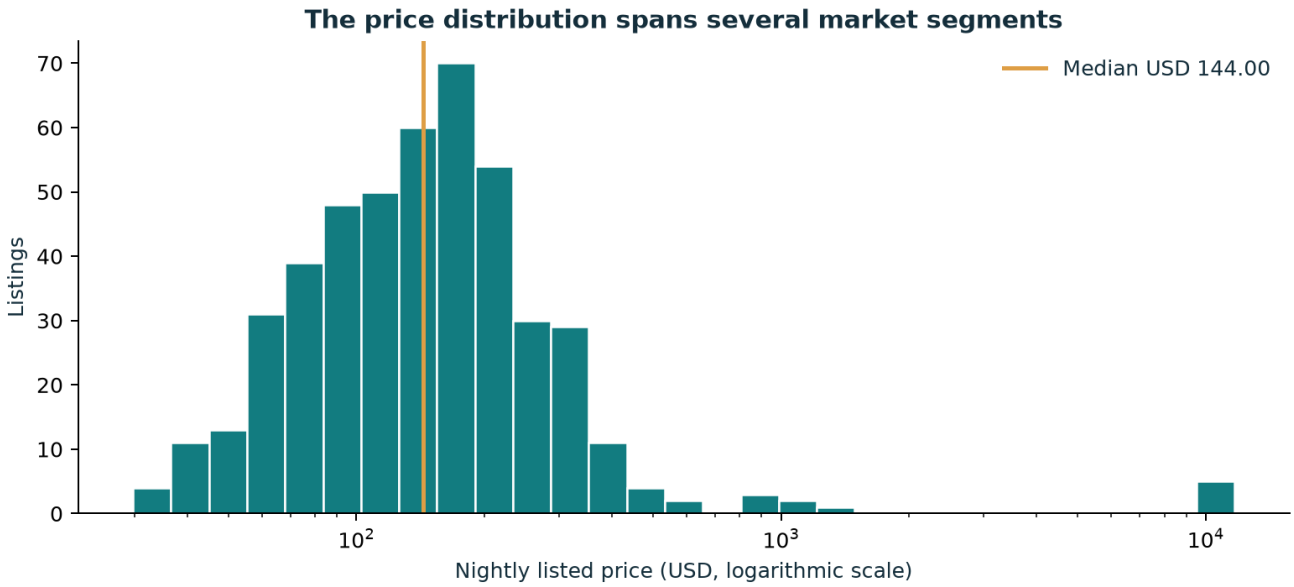


Figure 1. Positive prices only; outliers remain in both the underlying statistics and this log-scale histogram.

Across 467 priced listings, the median is $144.00; the 90th percentile is $308.00. The mean of $292.59 gives more weight to expensive units. Neither measure is a transaction-weighted average.

A prospective host or market analyst should start with a comparable set rather than the city-wide mean. Room type, capacity, property category and location describe materially different offerings. The explorer exposes room and ward filters for this reason.

Missing prices are not assigned the overall median. Doing so would mechanically narrow the distribution and exaggerate coverage. Outliers are retained because a high price may be valid for a large or unusual property.

Action: show the median, the number of priced listings and the number excluded together whenever a price benchmark is presented.

Room type and neighbourhood

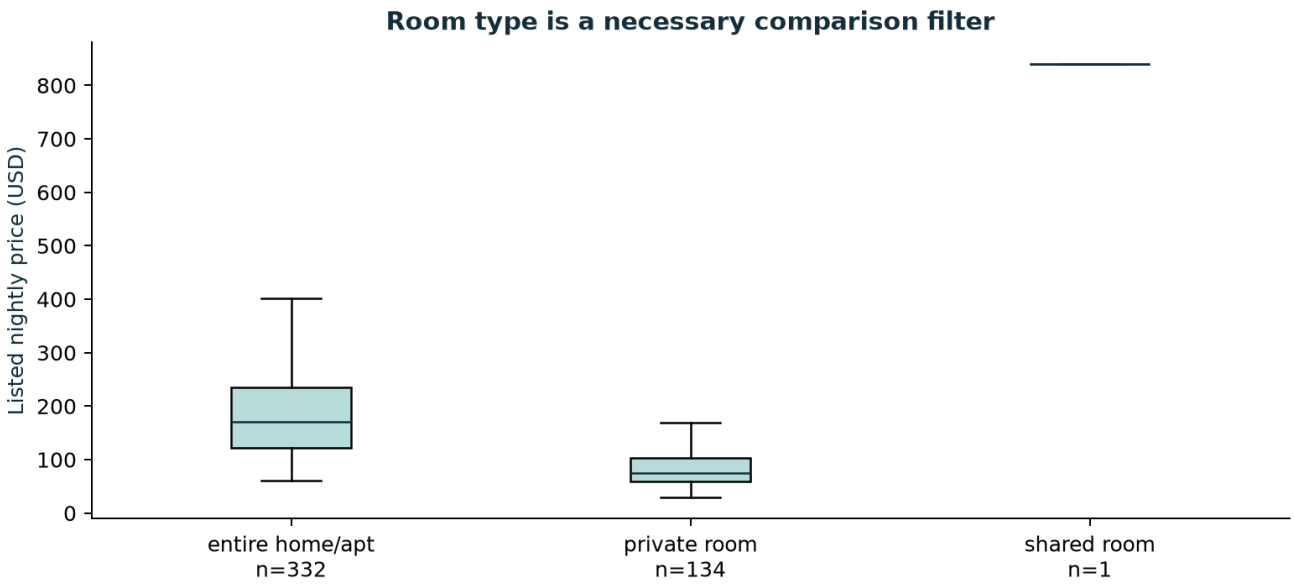


Figure 2. Box plots hide individual outlier markers for readability; no records are removed from analysis.

Room type	Listings	Priced	Median USD
entire home/apt	352	332	170.00
private room	147	134	75.00
shared room	1	1	840.00

Ward medians are published in neighbourhoods.csv, together with counts, rating summaries and confidence intervals for mean prices. Higher median prices may reflect a greater share of entire homes or larger units, so location should not be interpreted as the sole cause.

Confidence intervals resample hosts rather than individual listings. A selected host contributes all listings in that group, preserving within-host dependence. Groups with fewer than five hosts have no interval reported. Intervals are conditional on this observed snapshot, not guarantees about the whole rental market.

Action: use ward comparisons to form questions for deeper comparable-property research. Do not treat a small ward's rank as a stable investment ranking.

Geographic supply footprint

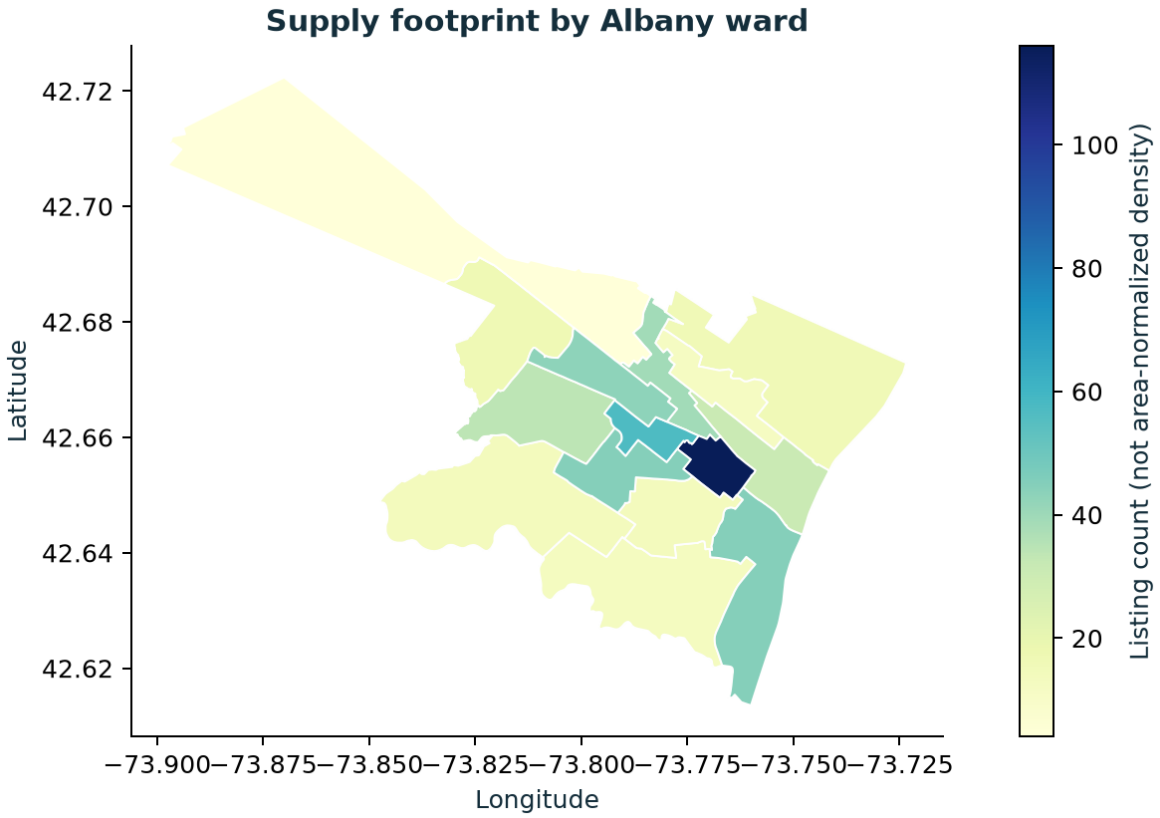


Figure 3. Official source ward boundaries coloured by listing count; not units per square kilometre.

The geographic view uses neighbourhood polygons rather than exact listing points. It describes the distribution of advertised supply across source-defined wards. 2 source features have null geometry and cannot be drawn; their listings remain in non-map summaries.

Counts are not area-normalized density: wards differ in area, land use and housing stock. Without a valid housing denominator, the map cannot tell us what share of local housing is used for short stays.

Action: identify wards for follow-up supply monitoring, then compare room-type composition and review activity. A concentration of advertisements is not by itself evidence of legal status, displacement or property-level profitability.

No external basemap, geocoding API or paid key is required. City-centre premiums are deferred because no defensible centre definition or travel-time model has been introduced.

RESEARCH & ENGINEERING

Host concentration and supply

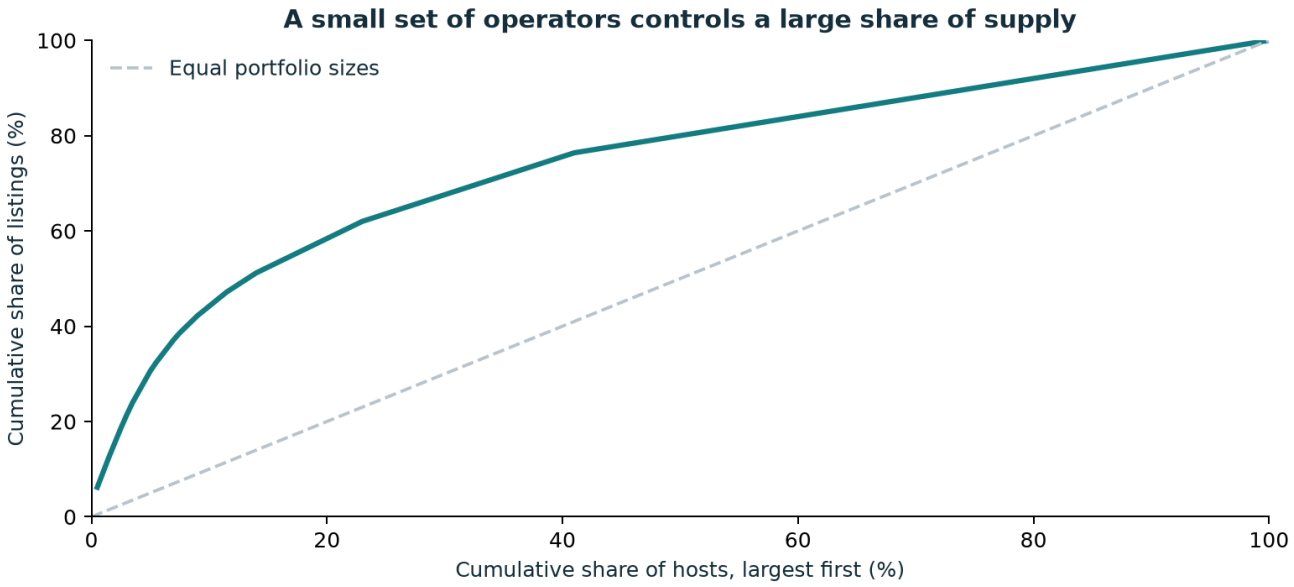


Figure 4. Hosts ordered from largest to smallest observed city portfolio.

The top 10% of 200 observed host accounts control 44.2% of the 500 listings. Portfolio size counts only listings in this city's snapshot and is separate from platform-reported global host counts.

The project labels accounts as single-listing or multiple-listing. Multiple listings can indicate professional activity, but this is a segmentation heuristic, not a verified commercial classification. A manager may act on behalf of several owners.

Action: monitor concentration across future snapshots and compare operator segments within room type. Large portfolios can affect aggregate pricing or availability without reflecting a market-wide change.

For model validation, all listings from the same account are kept on the same side of the train/test boundary. This makes the exercise closer to predicting prices for an unseen host and avoids an obvious source of overoptimistic evaluation.

RESEARCH & ENGINEERING

Forward availability and demand

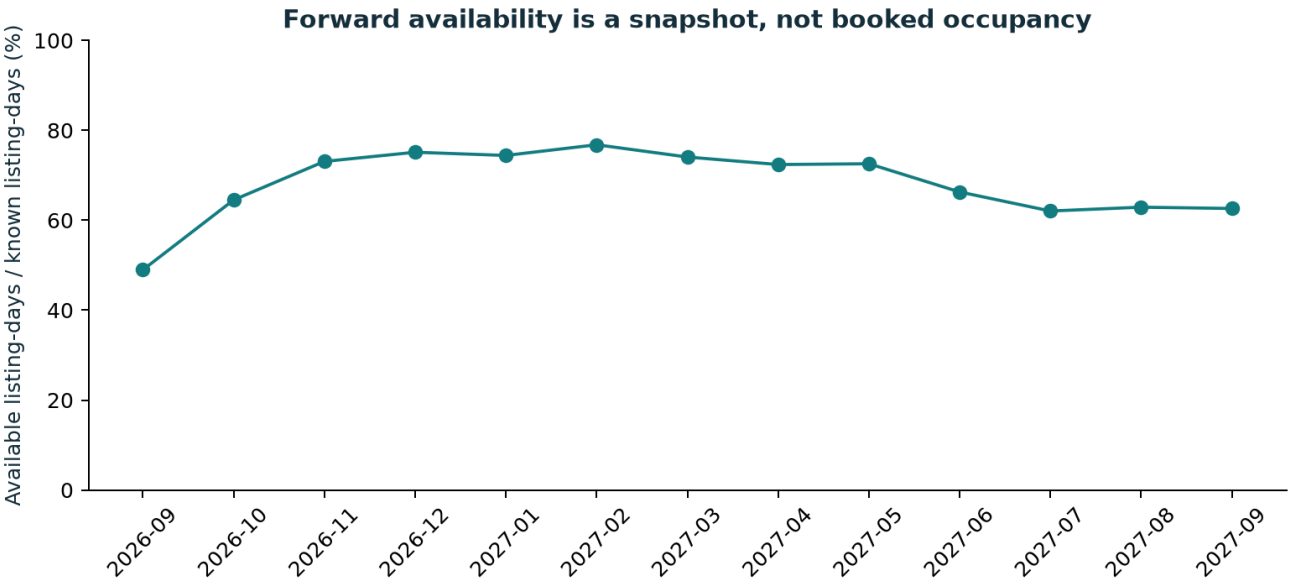


Figure 5. Monthly share of known calendar states that are available; edge months cover partial periods.

Each listing contributes 365-365 calendar observations. The chart is a forward-looking view from one scrape. Far-future dates may be unavailable because hosts have not opened their calendars; near-term dates may be blocked or booked.

The denominator is known availability states, not an unconditional 365. Unknown states remain missing. Monthly totals include partial first and last months, and therefore comparisons should use shares and coverage rather than raw counts alone.

Calendar nightly prices are missing in this snapshot. Weekend price testing, seasonal price curves and gross revenue scenarios are therefore explicitly unavailable. Repeating a listing's current price across 365 days would invent observations.

Action: collect repeated snapshots before studying changes in availability. Observing a transition still does not reveal whether a booking caused it.

Reviews and service signals

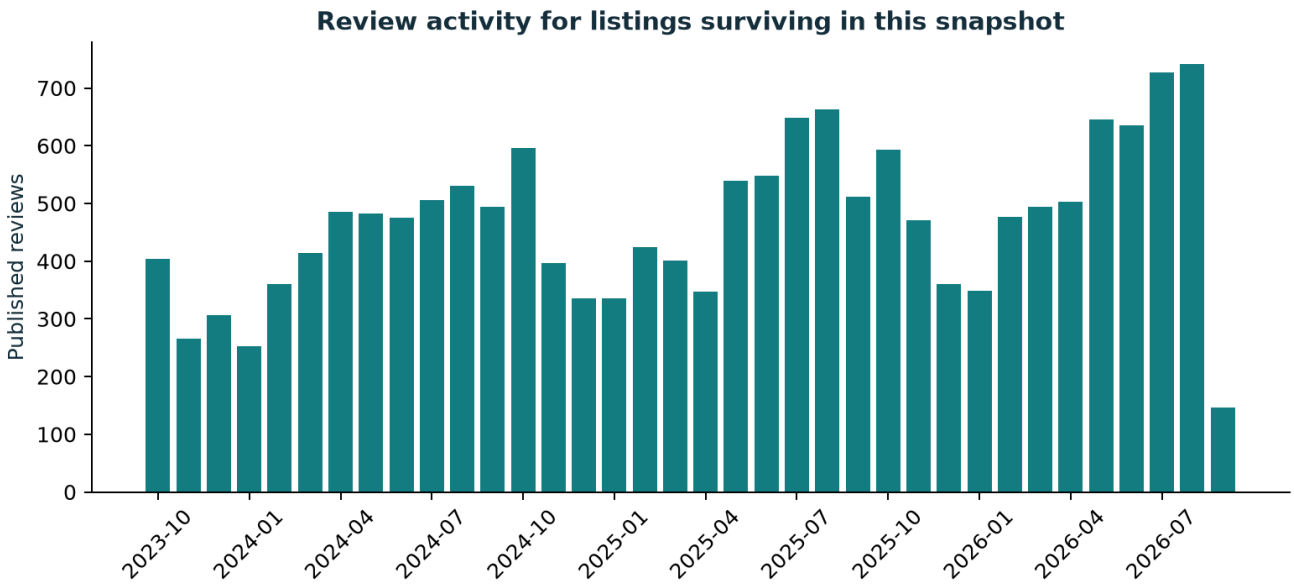


Figure 6. Last 36 observed review months for listings present in the current snapshot; final month is partial.

The retained review table contains 30,251 records. Recent-review frequency counts reviews in the 365 days ending at the observed scrape date. It is a demand-related proxy rather than a count of bookings or guests.

There are 449 listings with an overall rating; the unweighted mean is 4.74 out of five. High values compress differences and can make a small numerical improvement look more meaningful than it is.

Review publication can lag a stay. Listings removed before this snapshot are absent from the historical series, creating survivor bias. The final month is incomplete, so a decline at the end is not evidence that bookings are falling.

Action: combine recent-review volume, minimum evidence thresholds and rating subdimensions before investigating service issues. Review text analysis is deferred; no guest text is sent to an external AI service.

RESEARCH & ENGINEERING

Statistical design and findings

H1-H3 compare one deterministic representative listing per host, selected by string-sorted listing ID before filtering. This limits within-host dependence but changes the target from all listings to selected host representatives. Missing outcomes are removed per test. H3 excludes exactly ten reviews.

Question	Test	Holm p	Result
H1: Entire home vs private room mean nightly price	Welch independent t test	0.0334	Evidence against null
H2: Superhost vs non-superhost mean rating	Welch independent t test	0.0000	Evidence against null
H3: More than 10 vs fewer than 10 reviews: mean price	Welch independent t test	0.4647	Insufficient evidence
H4: Price distributions differ across neighbourhoods	Kruskal-Wallis	0.1751	Insufficient evidence
H5: Weekend minus weekday nightly price	One-sample t test of host-average paired differences	Not available	Not estimable

H1 and H2 use one-sided alternatives; H3 is two-sided. Welch tests do not assume equal variances. Shapiro-Wilk and median-centred Levene diagnostics are saved in analysis.json, alongside sample sizes, mean differences, two-sided 95% intervals and Cohen's d.

H4 uses Kruskal-Wallis on wards with at least five eligible hosts because price distributions are skewed. It tests distributional differences, not specifically equality of means. Its epsilon-squared effect size is reported in the machine-readable results. H5 requires within-listing weekday/weekend prices and then averages differences by host.

Holm adjustment covers the estimable hypothesis family. Failure to reject is not proof of equivalence. Statistical significance does not establish business importance, and no result is causal: property mix, selection and measurement limitations remain.

Because normality diagnostics indicate substantial departures, analysis.json also reports a 20%-trimmed Welch sensitivity test (Holm-adjusted across H1-H3), a rank test and probability-of-superiority effects. These target trimmed means/distributions, not the original arithmetic means. Compare conclusions rather than treating them as interchangeable proofs.

RESEARCH & ENGINEERING

Price associations and regression

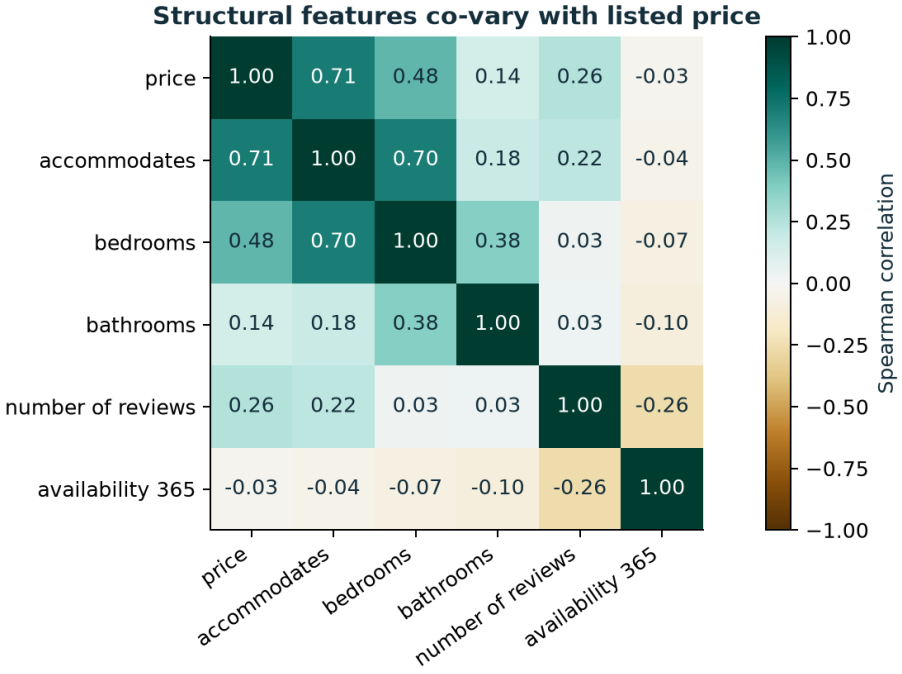


Figure 7. Pairwise Spearman correlations; missing pairs are omitted, so effective sample sizes vary.

The exploratory log-price OLS uses 467 complete cases and 184 host clusters. Its R-squared is 0.288. Predictors are capacity, bedrooms, bathrooms and host tenure; clustered standard errors allow errors within a host to be related.

Coefficients, confidence limits, p-values and variance inflation factors are saved in analysis.json. Bedroom and capacity effects may be unstable when those attributes move together. VIF is a diagnostic for this dependence, not an instruction to automatically delete a predictor.

This reduced model omits location and unobserved quality. A coefficient describes conditional association within the selected complete-case sample. It should not be presented as the price increase caused by adding a bedroom or earning another year of host tenure.

Action: use correlations and regression to identify questions and confounding, then judge predictive usefulness separately on held-out hosts.

RESEARCH & ENGINEERING

Predictive modeling protocol

The target is the positive published nightly listing price in USD. It is neither achieved booking revenue nor an optimal recommended price. All valid priced records, including expensive observations, remain eligible.

Stage	Design	Reason
Holdout	80 rows from 37 unseen hosts	One fixed 20% host split; seed 42.
Training	387 rows from 147 hosts	Group cross-validation keeps host accounts together.
Preparation	Fold-local numeric median and categorical mode; one-hot encoding	No fitting transformations on the holdout.
Comparison	Median baseline, Ridge, Extra Trees, histogram gradient boosting	Compare a simple reference and three model families.
Selection	Lowest mean training group-CV MAE	Holdout performance does not select the winner.
Reporting	MAE, RMSE and MAPE; residual slices	Different metrics expose different failure modes.

Features use structural and location attributes, amenity count, room/property type and account tenure. Price-per-bedroom, neighbourhood price aggregates, source revenue estimates and target-derived fields are deliberately excluded.

The fixed parameter choices are modest rather than extensively tuned. One seed and one city do not establish stable generalisation. No temporal split can be demonstrated with only one snapshot, and unseen-city behavior has not been tested.

Selected model: **median_baseline**. Selection criterion: training group-CV mean absolute error.

Model evidence and failure modes

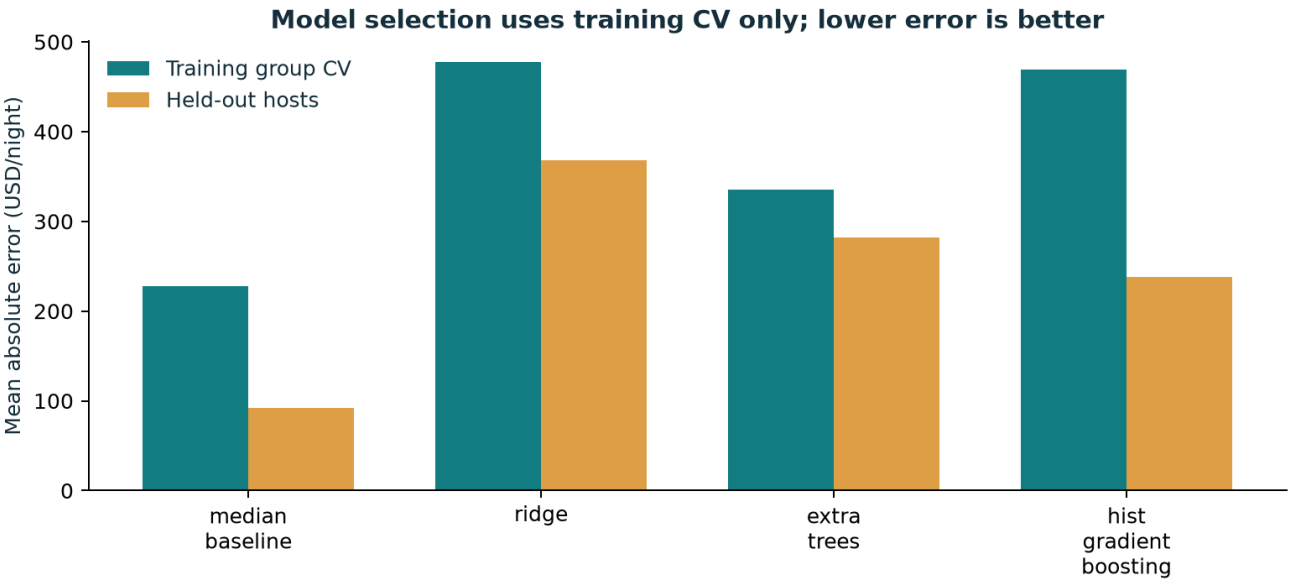


Figure 8. Training cross-validation and holdout errors shown separately. All scores are computed, not hand-entered.

Model	CV MAE	Test MAE	Test RMSE	MAPE %
median_baseline	227.40	92.46	130.23	84.61
ridge	477.25	367.82	472.20	303.59
extra_trees	335.49	282.23	772.55	214.33
hist_gradient_boosting	468.81	238.02	400.27	176.20

Permutation importance measures the increase in holdout MAE after shuffling one raw predictor. It is generated for the best non-baseline model by training CV, even if the median baseline is selected overall. Correlated features can share importance, and negative importance can occur through noise.

The residual figure and neighbourhood error table reveal uneven performance. Neighbourhood error slices with fewer than three test records are omitted. This is a diagnostic, not a fairness certification; small samples prevent reliable subgroup guarantees.

Action: compare any future model against the median baseline, repeat host splits, validate on a later snapshot and establish acceptable error by use case before deployment. A low average error alone is not sufficient for automatic pricing.

RESEARCH & ENGINEERING

Recommendations and roadmap

Priority	Recommendation	Evidence / prerequisite
Now	Publish segmented benchmarks with coverage	Real prices, missingness and room/ward composition are observable.
Now	Retain quality and source manifests	Hashes and row counts make refreshes auditable.
Next	Add a second dated snapshot	Test change tracking, survivor effects and model stability.
Next	Add robust statistical sensitivity checks	Skew and representative-host selection can influence inference.
Later	Scale ingestion by city/date partitions	Add job orchestration, locks, failure alerts and bounded concurrency.
Later	Consider NLP or AI assistance	Need language checks, privacy review and an evaluation set first.

For 50 or more cities, store immutable raw objects and partition analytical data by city, scrape date and calendar month. Use a job queue with per-city retries and a centralized metadata catalog. Avoid materializing every full review corpus in pandas memory.

A cloud design could pair object storage with scheduled container jobs and a query engine, but no paid resources were provisioned. Cost, latency and freshness targets must be agreed before selecting a managed platform.

Open limitations include absent calendar prices, incomplete price coverage, source selection/survivor bias, uncertain calendar semantics and only one validated market. Row-level CDC, true bookings/revenue, causal inference, NLP/RAG, live deployment and automated business recommendations are not claimed as completed.

Portfolio focus: explain why a missing capability remains missing. A transparent boundary is more useful than a confident number unsupported by the source.

TRANSPARENCY & NEXT STEPS

Ownership, learning and sources

AI assistance. OpenAI Codex assisted with project design, Python and SQL implementation, tests, documentation and generated-report narrative. The exact underlying model version is not exposed in this artifact. No claim is made that the code was independently written by the project owner.

The key user direction was to treat the brief as a personal career-development project and build it with clear heads-ups. Validation includes actual source downloads, warehouse assertions, synthetic edge-case tests, end-to-end execution and rendered artifact inspection. See docs/verification.md for the final executed checks.

Meaningful corrections include retaining large IDs as strings, accounting for the archive/scrape date mismatch, declining to invent missing calendar prices, grouping model validation by host and retaining fuzzy duplicate candidates rather than deleting them.

Learning ownership. Before using this work in an interview, run the pipeline yourself, change a quality rule, inspect a SQL join and reproduce one figure. Describe which parts were AI-assisted and which extensions you personally implemented. docs/career-guide.md provides a walkthrough and honest CV wording.

Reference	Use
insideairbnb.com/get-the-data/	Dated source downloads; CC BY 4.0 attribution.
insideairbnb.com/data-assumptions/	Calendar and geographic interpretation caveats.
Inside Airbnb Data Dictionary (linked in README)	Source field meanings; currency and account-tenure interpretation.
scikit-learn: GroupShuffleSplit documentation	Group-separated validation semantics.
SciPy: scipy.stats.ttest_ind documentation	Welch and directional test implementation.

Source files remain local and ignored by Git. Public outputs are aggregated or exclude identifiers and review text. The original project brief is not redistributed. Findings describe this frozen snapshot, not the current market.